APPENDIX 3: PAPUA NEW GUINEA—DEBT SUSTAINABILITY ANALYSIS[1]

Papua New Guinea (PNG) has reduced its risk of public debt distress from moderate to low since the 2011 Article IV Consultation. This reflects both PNG's reduction in public external debt over the past decade, and an enhanced debt capacity under the DSA following an upgrade of the country's policy and institutional rating by the World Bank[2]. Under the baseline scenario, all external debt sustainability indicators remain well below their applicable thresholds. Stress tests indicate that unstable debt dynamics only occur if prices of copper, gold and natural gas collapse and remain low over a decade, and the government borrows abroad to finance current expenditure plans. Total public debt continues its downward path under the baseline, and assets are accumulated after domestic debt is paid off in 2020. In a low risk but high impact scenario, cancellation of the LNG project would derail progress in debt reduction if current expenditure plans are maintained.

[1] Since Papua New Guinea is an IBRD/IDA blend country, this DSA is prepared by Fund staff in consultation with the World Bank and the Asian Development Bank (AsDB) under the IMF-WB DSA framework for Low-Income Countries. The fiscal year of Papua New Guinea is the calendar year.

[2] The latest Country Policy and Institutional Assessment (CPIA) rating for PNG was 3.30 in 2010 and the average over 2008-10 was 3.27, leading to a policy performance rating upgrade from poor (CPIA <= 3.25) to medium (3.25 < CPIA < 3.75).

Background

1. PNG has successfully reduced its public and publicly guaranteed (PPG) debt burden over the past decade. Total public debt declined from 71 to 25 percent of GDP from 2001 to 2011 (Table III.1). External public debt fell from 50 to 10 percent of GDP during this period (Tables III.1 and III.3). Multilateral lenders such as the World Bank and the AsDB account for about 75 percent of PPG external debt, and bilateral creditors account for the bulk of the remainder.

2. Private external debt has risen sharply in the run-up to LNG construction (Table III.3)**.** Private external debt jumped to an estimated 82 percent of GDP in 2011 from 12 percent of GDP in 2008,[3] driven by medium- and long-term debt draw-downs by the mineral and gas sectors and mainly related to the LNG project (construction is being undertaken by Esso Highlands, an ExxonMobil subsidiary).

[3] Staff apportioned the funds flowing into PNG to finance the LNG project in line with Esso Highlands' financing plans (30 percent equity, 70 percent debt).

Economic Outlook and Underlying DSA Assumptions

3. Until full scale LNG production in 2015, PNG's economic performance will be driven by LNG construction, commodity price developments, and spending associated with the 2012 elections. Real GDP growth increased to 9 percent in 2011 as LNG construction approached its peak phase, and growth is projected to remain high at 8 percent in 2012 before slowing to 4 percent in 2013. Large current account deficits (about 30 percent of GDP) during 2010–12 are being mostly financed by a combination of FDI and loan drawdowns by the mineral sector, and private sector external debt is expected to peak at 96 percent of GDP in 2013. The European debt crisis led to fluctuation in commodity prices in 2011, and although the global outlook has improved, risks remain. PNG's current account is most sensitive to the price of gold, while mineral tax revenues for the government derive mostly from the copper and oil sectors before 2015.

4. From 2015, real GDP and the current account will move in line with the LNG sector. LNG production will push up real GDP by 20 percent in 2015, and generate a persistent current account surplus. LNG revenues will be used to pay down the mineral sector's external debt and provide dividend payments to shareholders located abroad and in PNG. Fiscal revenues from LNG are projected to materialize from 2018 and total mineral revenues will reach around 10 percent of non-mineral GDP by 2024 (between the low and moderate revenue scenarios identified in FAD's technical assistance report for PNG's sovereign wealth fund). Box III.1 contains the medium-term macroeconomic framework for the DSA.

External Debt Sustainability Analysis

5. In the baseline scenario, all PPG external debt and debt service indicators stay well below the policy-dependent debt burden thresholds (Figure III.1 and Table III.3)**.** The PV of PPG external debt as a percentage of GDP is on a declining path and stays under 10 percent in the medium term, far below the 40 percent threshold. The PVs of PPG external debt-to-exports and debt-to-revenue ratios are also expected to stay below their applicable thresholds. After 2015, the external debt burden is expected to fall even faster due to the large projected increase in GDP, exports and revenue from LNG production.

Policy-based PPG External Debt Burden Thresholds for PNG

	Thresholds	2011	PNG's ratios 2012–17	2018–32
PV of debt in percent of				
GDP	40	8.6	7.2	4.8
Exports	150	14.5	13.1	10.5
Revenues	250	31.4	33.8	23.0
Debt service in percent of				
Exports	20	1.3	1.1	0.8
Revenues	20	2.7	2.7	1.7

6. Public external debt sustainability is maintained under all standard stress tests, and would be threatened only in a staff-constructed scenario with a very poor fiscal policy response to a substantial terms of trade shock (Table III.4)**.** Consistent with WEO commodity price forecasts, the baseline scenario predicts PNG's terms of trade to improve in 2012 before settling at a permanently high level. This, combined with the boost from LNG production, makes PNG resilient to a wide range of GDP growth and depreciation shocks. Staff can only generate unstable public external debt dynamics if we instead use the PNG authorities' conservative estimates for export prices for the post-2013 period, impose a permanent reversal of the WEO-projected 16 percent gas price increase between 2011 and 2012,[4] and then combine these developments with a government decision to borrow extensively abroad to finance the baseline levels of real consumption and imports, instead of allowing economic adjustment to the terms of trade change. Even in this very low probability scenario, the PV of PPG external debt breaches the threshold of 40 percent of GDP only in 2018. Until 2015, the prices of gold and copper are the most important prices for external sustainability; thereafter, the LNG price is the dominant factor.

7. Delays to the LNG project would generate adverse public and private debt dynamics, even if PPG external debt indicators remain within the thresholds. The historical scenario graphs in Figure III.1 can be interpreted as representing PNG's external public debt dynamics in the absence of LNG construction and production, assuming that historical levels of real GDP growth are maintained. As for private sector external debt, staff projects that when LNG production begins, this debt will be paid down, falling to 21 percent of GDP by 2024. Delays to the

[4]The authorities' projected prices for gold, copper and oil are used. Together with the imposed gas price decline, this amounts to a substantial negative shock to the mineral balance relative to the baseline, growing to 12 percentage points of GDP by 2016 and remaining at that level afterward.

project, or a fall in LNG prices, would set back the private sector amortization schedule envisaged in the DSA.

Public Debt Sustainability Analysis[5]

8. Over the medium and long term, staff's baseline projections for PNG show a full pay-down of domestic public debt by 2020, followed by asset accumulation (Figure III.2 and Table III.1)**.** The projected downward trend in public debt is aided by the jump in real GDP owing to full-scale LNG production from 2015. Under the authorities' budget assumptions, the transitional period of low mineral revenues during 2013–17 is associated with low government expenditure, so any deficits remain small and the PV of public sector debt continuously declines from 24 percent of GDP in 2011 to 11 percent of GDP in 2017. From 2018 onwards, large projected LNG-related revenues enter the budget without a commensurate increase in government expenditure, resulting in long-term surpluses of about 4 percent of GDP. Domestic public debt is paid off in 2020 and assets are accumulated in a sovereign wealth fund, which measures 29 percent of GDP by 2032.

9. Economic growth and saving of commodity tax revenues are the key factors for the projected public debt path as shown in alternative scenarios and bound tests (Figure III.2 and Table III.2)**.** The PV of public debt to GDP would remain at 11 percent by 2032 if there is a temporary negative shock to real GDP growth in 2013-14 followed by permanently lower revenues and unchanged expenditures.[6] Such a scenario would likely not materialize because the authorities would adjust fiscal policy. Keeping the primary fiscal deficit at zero from 2012 through 2032, instead of moving into large surpluses over the medium term as in the baseline, would result in the PV of public debt being at 13 percent of GDP by 2017 and rising to 46 percent of GDP by 2032. This scenario is unrealistic but quantifies the sensitivity of the public debt outlook to the LNG revenue projections.

10. Realization of non-contingent liabilities related to superannuation arrears and SOEs, together with the cancellation of the LNG project, would generate unstable debt dynamics (Figure III.2 and Table III.2)**.** The government's unfunded superannuation liabilities are estimated to be 7 percent of GDP by end-2011, and staff assumes that SOEs' debt-to-GDP ratio is equal to the 2.5 percent of GDP estimated in 2008.[7] The government has issued a completion guarantee for the LNG

[5] Public debt includes domestic and external debt issued by the central government, less assets in the sovereign wealth fund.

[6] Growth rates during 2013–14 are set to 2 percent, the historical average minus one standard deviation.

[7] This estimate in the 2008 Article IV DSA was based on end-2007 financial statements of nine SOEs. Off-balance sheet liabilities, which may be significant, were not considered. Since 2008, data updates have been unavailable.

project of K 5.4 billion, amounting to 18 percent of GDP in 2011, and the public balance sheet would have to absorb this loss if the project is abandoned. Realizations of the above liabilities would not of themselves derail the downward trend in public debt ratios, but the loss of LNG revenues would put public debt on an upward path under current expenditure plans (to a PV of 33 percent of GDP by 2032).

Conclusion

11. Since the 2011 SR, PNG has successfully reduced its risk of public external debt distress from medium to low, and the commendable public debt performance is projected to continue. The authorities' spending plans indicate sustained reduction in both domestic and external public debt over the medium and long term, even during the 2013–17 period of transitionally low fiscal revenues. For expositional purposes in this report, it is assumed that domestic debt is fully paid down first and after that, assets accumulated in a sovereign wealth fund.

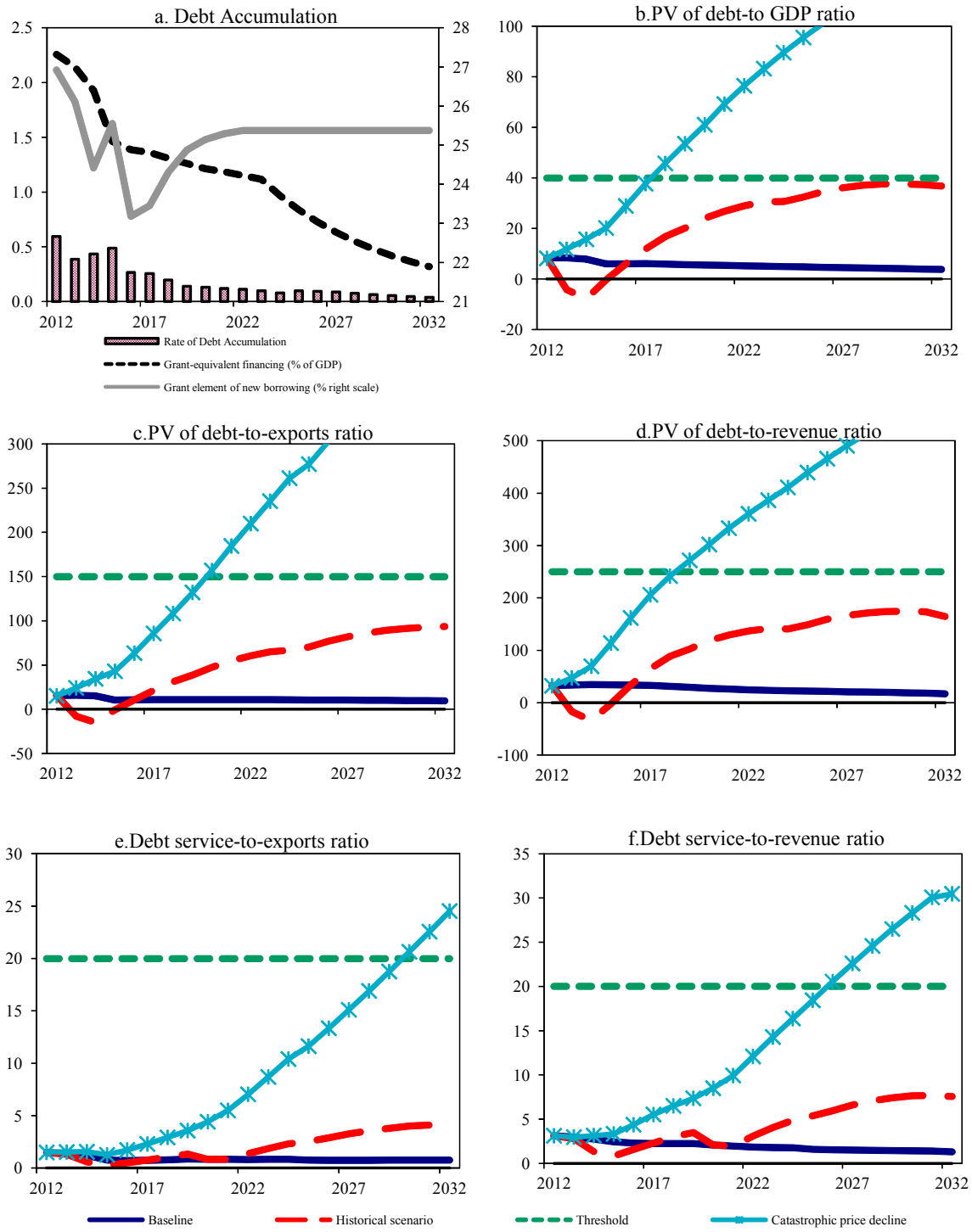
12. PNG's debt dynamics remain stable under standard stress-testing, and would only become unstable in the aftermath of low probability catastrophic shocks. Staff calculate that a severe and permanent shock to gold, copper, oil and gas prices, coupled with a loss of fiscal discipline and extensive foreign borrowing to maintain consumption and imports at baseline levels, would generate an unsustainable debt path. Realization of non-contingent liabilities related to superannuation arrears and SOEs would delay any progress in debt reduction. In the low risk but high impact scenario where the LNG project is cancelled, loss of associated fiscal revenues would ratchet up public debt, assuming that current expenditure plans remain unchanged. In that event, staff would advise that expenditure plans be curbed sharply downward.

Box III.1. Macroeconomic Assumptions Underlying the DSA

- **Real GDP growth** is projected to be 8 percent on average over the medium term, above the historical average of 4 percent, and slow gradually to 3–4 percent in the long run. Over the medium term, growth in the LNG and non-mineral sectors will offset the decline in copper and petroleum production.
- Construction of the **LNG project** will pass its peak phase in 2013, and be finished by 2014. Production and exports are expected to start in 2014 and reach full capacity in 2015, with a maximum capacity of 6.6 million tons of LNG produced annually. Staff estimates that LNG production will raise the level of real GDP by about 20 percent in 2015. However, accounting for substantial income outflows, the LNG project is expected to increase annual GNI by about 8 percent.
- **Inflation** is projected to decline to 7 percent in the near term due to the end of PNG production. It will stabilize at 6.5 percent by 2017, and at around 5 percent in the long run (after 2021).
- The **current account** is in deficit in 2011 and the deficit is expected to remain sizeable until 2015, reflecting strong import growth as well as services and income deficits associated with LNG construction. FDI and medium- to long-term debt inflows are financing most of the imports.
- The **grant element of loans** is expected to decline. As GDP per head rises, the share of external financing provided on concessional terms is expected to decline slightly over the projection period.[1/]
- The **primary fiscal balance** is estimated to be in surplus of 1.5 percent of GDP in 2011, accounting for spending from the trust accounts. Over the medium term small primary deficits are anticipated, but these turn into large surpluses especially after 2022, as LNG tax revenues are realized.

1/ Grant-equivalent financing (in percent of GDP) decreases over time. From 2016 onwards, the majority of new AsDB disbursements are through OCR facilities, which carry a higher interest rate than ADF loans.

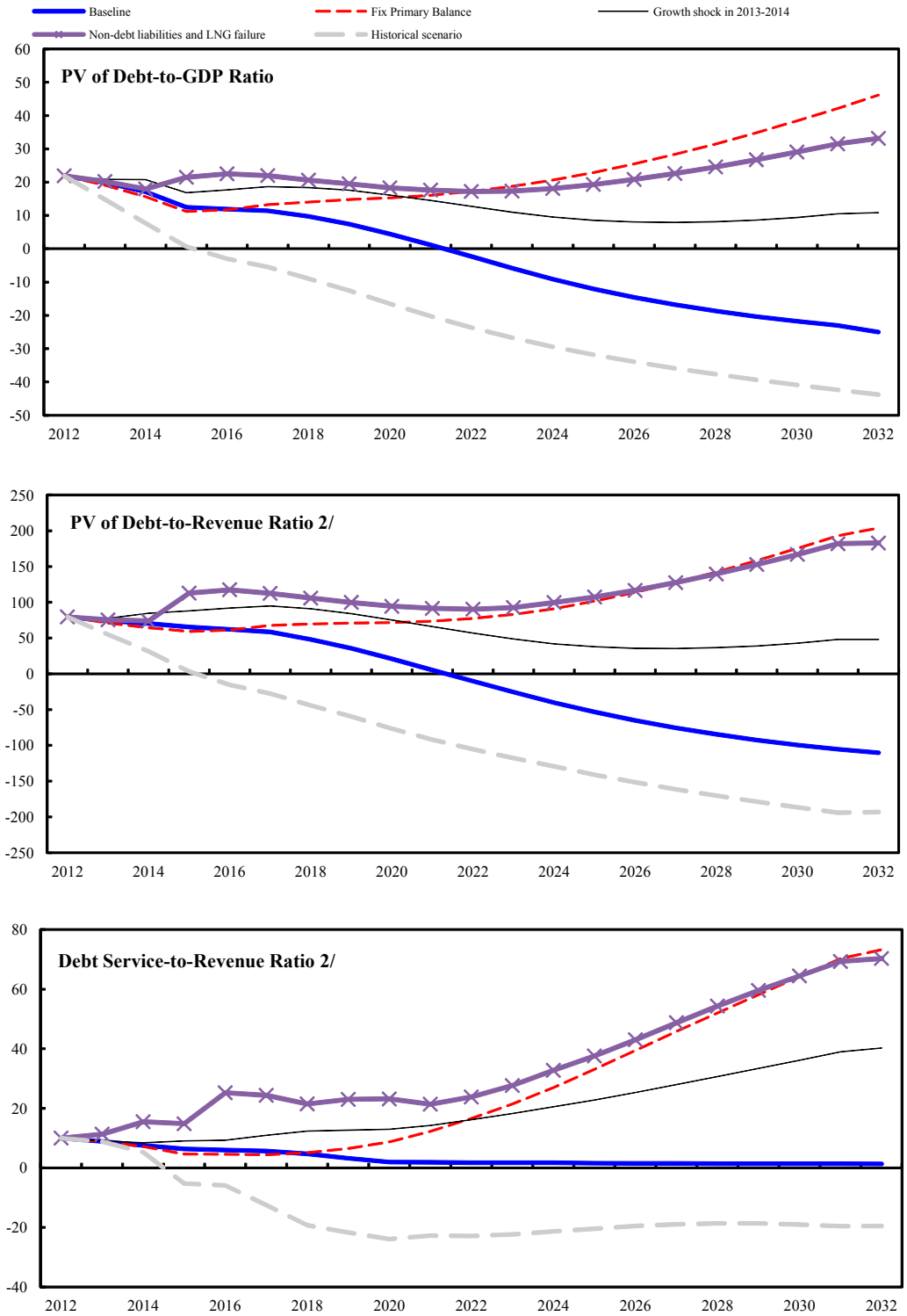
Figure III.1. Papua New Guinea: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2012-2032 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2022. In all figures it corresponds to a catastrophic price decline shock.

Figure III.2. Papua New Guinea: Indicators of Public Debt Under Alternative Scenarios, 2012-2032 1/



Sources: Country authorities; and staff estimates and projections.

1/Gross domestic and external central government debt less assets in sovereign wealth fund.
The most extreme stress test is the test that yields the highest ratio in 2022.
In all figures this corresponds to the failure of the LNG plant.

2/ Revenues are defined inclusive of grants.

Table III.1. Papua New Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2009-2032
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2009	2010	2011	Average 5/	Standard Deviation 5/	2012	2013	2014	2015	2016	2017	2012-17 Average	2022	2032	2018-32 Average
Public sector debt 1/	31.5	25.6	25.2			23.2	21.3	18.4	13.5	13.0	12.5		-1.4	-24.3	
o/w foreign-currency denominated	12.5	10.4	9.8			9.7	9.8	9.2	7.1	7.1	7.1		6.2	4.5	
Change in public sector debt	-0.2	-5.9	-0.4			-2.0	-1.9	-2.9	-4.9	-0.6	-0.5		-3.5	-2.0	
Identified debt-creating flows	8.7	-8.2	-5.3			-0.2	0.2	-1.2	-3.9	-1.4	-2.3		-4.0	-2.5	
Primary deficit	7.6	-4.4	-1.5	-3.8	5.1	1.3	0.9	0.6	0.0	-1.0	-1.9	0.0	-4.1	-4.4	-3.8
Revenue and grants	27.3	31.3	29.8			27.5	26.8	24.3	19.0	19.2	19.6		22.3	22.6	
of which: grants	3.9	5.3	2.5			1.9	1.9	1.7	1.3	1.3	1.2		1.0	0.3	
Primary (noninterest) expenditure	34.9	26.9	28.3			28.8	27.7	24.9	19.0	18.2	17.6		18.1	18.3	
Automatic debt dynamics	1.1	-3.8	-3.8			-1.5	-0.7	-1.8	-3.9	-0.4	-0.3		0.1	1.9	
Contribution from interest rate/growth differential	0.5	-2.6	-1.9			-1.5	-0.7	-1.8	-3.9	-0.4	-0.3		0.1	1.9	
of which: contribution from average real interest rate	2.3	-0.4	0.2			0.2	0.2	-0.3	-0.8	0.3	0.3		0.2	1.3	
of which: contribution from real GDP growth	-1.8	-2.2	-2.1			-1.8	-0.9	-1.5	-3.1	-0.7	-0.6		-0.1	0.6	
Contribution from real exchange rate depreciation	0.6	-1.2	-1.9			0.0	0.0	0.0	0.0	0.0	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-8.9	2.3	4.8			-1.7	-2.1	-1.7	-1.0	0.8	1.8		0.5	0.5	
Other Sustainability Indicators															
PV of public sector debt	...	...	24.0			21.9	19.9	17.1	12.5	11.9	11.4		-2.3	-25.0	
o/w foreign-currency denominated	...	...	8.6			8.4	8.4	7.9	6.1	6.1	6.1		5.3	3.8	
o/w external	...	...	8.6			8.4	8.4	7.9	6.1	6.1	6.1		5.3	3.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	17.6	4.5	5.0			8.4	5.9	3.9	1.9	0.8	-0.2		-3.7	-4.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	80.4			79.7	74.4	70.4	65.6	62.2	58.5		-10.3	-110.4	
PV of public sector debt-to-revenue ratio (in percent)	...	...	87.7			85.6	79.9	75.7	70.3	66.6	62.4		-10.8	-111.7	
o/w external 3/	...	...	31.4			32.9	33.8	34.8	34.2	33.9	33.2		24.8	17.0	
Debt service-to-revenue and grants ratio (in percent) 4/	14.4	9.7	8.8			10.0	8.9	7.6	6.4	6.0	5.7		1.8	1.3	
Debt service-to-revenue ratio (in percent) 4/	16.9	11.7	9.6			10.8	9.5	8.2	6.9	6.4	6.0		1.9	1.3	
Primary deficit that stabilizes the debt-to-GDP ratio	7.8	1.5	-1.1			3.3	2.8	3.5	4.9	-0.4	-1.5		-0.6	-2.4	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.1	7.6	8.9	5.0	2.7	7.7	4.0	7.7	20.0	5.4	4.9	8.3	3.0	3.0	3.2
Average nominal interest rate on forex debt (in percent)	2.0	1.6	1.6	2.6	0.7	1.6	1.7	1.8	1.9	2.0	2.0	1.8	2.1	2.2	2.1
Average real interest rate on domestic debt (in percent) 6/	12.7	-2.5	1.7	3.1	5.8	2.0	1.7	-2.4	-10.4	4.6	5.2	0.1	...	...	-0.4
Real exchange rate depreciation (in percent, + indicates depreciation)	4.7	-10.1	-19.8	-8.4	7.6	0.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent) 6/	-2.6	9.9	4.9	6.1	4.4	5.6	5.9	10.3	20.4	3.8	3.6	8.3	5.0	5.0	5.3
Growth of real primary spending (deflated by GDP deflator, in percent)	0.4	-0.2	0.1	0.1	0.2	0.1	0.0	0.0	-0.1	0.0	0.0	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	26.9	26.1	24.4	25.6	23.2	23.4	24.9	25.4	25.4	...

Sources: Country authorities; and staff estimates and projections.
1/ Gross debt of central government less assets in sovereign wealth fund. State-owned enterprises are not included due to data limitations. Deposits held at the Bank of PNG are not included.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
6/ The GDP deflator increases sharply in 2014-15 owing to a base year effect as the share of real activity accounted for by the oil and gas sector (which has experienced high inflation since the base year of 1998) increases sharply. Rebasing the national accounts to a more recent year is recommended. This would push up projected real GDP growth in 2014-15, reduce the GDP deflator and increase the imputed average real interest rates.

Table III. 2: Sensitivity Analysis for Key Indicators of Public Debt, 2012–2032
(In percent)

	Projections							
	2012	2013	2014	2015	2016	2017	2022	2032
PV of Debt-to-GDP Ratio								
Baseline	22	20	17	12	12	11	-2	-25
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	22	15	8	1	-3	-5	-24	-44
A2. Primary balance is fixed to zero	22	19	16	11	12	13	17	46
A3. Permanently lower GDP growth 1/	22	20	18	13	13	13	4	2
A4. Superannuation arrears, SOEs and LNG project cancellation	22	20	18	21	23	22	17	33
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	22	21	21	17	18	19	13	11
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	22	20	18	14	13	13	-1	-24
B3. Combination of B1-B2 using one half standard deviation shocks	22	18	14	10	11	11	2	-8
B4. One-time 30 percent real depreciation in 2013	22	24	21	16	15	14	1	-21
B5. 10 percent of GDP increase in other debt-creating flows in 2013	22	30	27	21	21	20	7	-13
PV of Debt-to-Revenue Ratio 2/								
Baseline	80	74	70	66	62	59	-10	-110
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	80	55	31	3	-15	-27	-106	-193
A2. Primary balance is fixed to zero	80	71	64	59	61	68	77	204
A3. Permanently lower GDP growth 1/	80	75	73	70	69	68	17	8
A4. Superannuation arrears, SOEs and LNG project cancellation	80	75	74	113	118	112	90	183
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	80	78	85	88	92	95	57	48
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	80	76	76	71	68	64	-5	-104
B3. Combination of B1-B2 using one half standard deviation shocks	80	66	56	54	55	56	9	-36
B4. One-time 30 percent real depreciation in 2013	80	88	85	82	77	73	3	-91
B5. 10 percent of GDP increase in other debt-creating flows in 2013	80	113	112	112	108	104	33	-59
Debt Service-to-Revenue Ratio 2/								
Baseline	10	9	8	6	6	6	2	1
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	10	9	5	-5	-6	-12	-23	-19
A2. Primary balance is fixed to zero	10	9	7	5	5	4	17	73
A3. Permanently lower GDP growth 1/	10	9	8	7	7	7	7	30
A4. Superannuation arrears, SOEs and LNG project cancellation	10	11	15	15	25	24	24	70
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	10	9	8	9	9	11	16	40
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	10	9	8	8	7	7	3	3
B3. Combination of B1-B2 using one half standard deviation shocks	10	9	7	2	2	2	5	20
B4. One-time 30 percent real depreciation in 2013	10	10	9	8	8	8	5	7
B5. 10 percent of GDP increase in other debt-creating flows in 2013	10	9	13	21	16	18	13	14

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table III.3: External Debt Sustainability Framework, Baseline Scenario, 2009-2032 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2009	2010	2011			2012	2013	2014	2015	2016	2017	2012-2017 Average	2022	2032	2018-2032 Average
External debt (nominal) 1/	**55.0**	**67.2**	**91.9**			**95.0**	**105.6**	**87.9**	**60.6**	**56.4**	**52.4**		**32.4**	**27.5**	
o/w public and publicly guaranteed (PPG)	12.5	10.4	9.8			9.7	9.8	9.2	7.1	7.1	7.1		6.2	4.5	
Change in external debt	30.1	12.2	24.7			3.1	10.6	-17.7	-27.3	-4.1	-4.1		-3.2	-0.1	
Identified net debt-creating flows	10.8	7.0	7.9			14.8	10.4	-4.7	-23.6	-13.9	-12.7		-8.9	-6.9	
Non-interest current account deficit	**15.2**	**24.4**	**35.4**	**3.6**	**16.1**	**26.9**	**18.5**	**11.2**	**-10.5**	**-10.2**	**-10.4**	**4.3**	**-8.5**	**-7.3**	**-8.5**
Deficit in balance of goods and services	9.2	19.4	20.2			17.0	9.6	3.2	-25.8	-27.1	-25.9		-21.4	-17.2	
Exports	57.9	62.3	59.2			54.0	53.3	51.9	57.7	57.2	55.7		48.0	39.4	
Imports	67.1	81.7	79.4			71.0	62.9	55.2	31.9	30.1	29.7		26.7	22.2	
Net current transfers (negative = inflow)	-2.1	-2.0	-1.2	-4.1	2.5	-0.7	-0.6	-0.4	-0.2	0.0	-0.3	-0.4	-1.0	0.2	-0.5
o/w official	-3.3	-2.8	-2.5			-1.9	-1.9	-1.7	-1.3	-1.3	-1.2		-1.0	-0.3	
Other current account flows (negative = net inflow)	8.1	6.9	16.5			10.6	9.5	8.4	15.5	16.9	15.8		13.8	9.8	
Net FDI (negative = inflow)	**-5.2**	**-8.7**	**-13.8**	**-4.3**	**4.5**	**-7.8**	**-6.1**	**-10.5**	**-1.5**	**-1.7**	**-1.7**	-4.9	**-1.5**	**-1.2**	-1.4
Endogenous debt dynamics 2/	**0.8**	**-8.7**	**-13.8**			**-4.3**	**-1.9**	**-5.5**	**-11.7**	**-2.0**	**-0.6**		**1.1**	**1.6**	
Contribution from nominal interest rate	1.2	1.3	1.0			1.5	1.7	1.8	1.1	1.1	2.0		2.2	2.3	
Contribution from real GDP growth	-1.5	-3.4	-4.7			-5.8	-3.6	-7.2	-12.8	-3.1	-2.7		-1.0	-0.8	
Contribution from price and exchange rate changes	1.2	-6.5	-10.0			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**19.2**	**5.2**	**16.9**			**-11.7**	**0.2**	**-13.0**	**-3.7**	**9.8**	**8.6**		**5.6**	**6.8**	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	90.7			93.7	104.2	86.6	59.5	55.4	51.3		31.5	26.8	
In percent of exports	...	...	153.3			173.4	195.4	166.7	103.1	96.8	92.3		65.6	68.1	
PV of PPG external debt	**...**	**...**	**8.6**			**8.4**	**8.4**	**7.9**	**6.1**	**6.1**	**6.1**	**7.2**	**5.3**	**3.8**	**4.8**
In percent of exports	**...**	**...**	**14.5**			**15.6**	**15.8**	**15.1**	**10.5**	**10.6**	**10.9**	**13.1**	**11.0**	**9.7**	**10.5**
In percent of government revenues	**...**	**...**	**31.4**			**32.9**	**33.8**	**34.8**	**34.2**	**33.9**	**33.2**	**33.8**	**24.8**	**17.0**	**23.0**
Debt service-to-exports ratio (in percent)	**13.4**	**14.7**	**15.7**			**16.5**	**15.3**	**31.0**	**19.6**	**17.0**	**19.9**		**22.7**	**23.5**	
PPG debt service-to-exports ratio (in percent)	**1.8**	**1.4**	**1.3**			**1.5**	**1.4**	**1.2**	**0.8**	**0.7**	**0.7**	**1.1**	**0.8**	**0.7**	**0.8**
PPG debt service-to-revenue ratio (in percent)	**4.4**	**3.3**	**2.7**			**3.1**	**3.0**	**2.8**	**2.4**	**2.3**	**2.3**	**2.7**	**1.9**	**1.3**	**1.7**
Total gross financing need (Billions of U.S. dollars)	1.4	2.5	3.9			4.3	3.3	3.1	-0.2	-0.6	-0.3		0.3	0.4	
Non-interest current account deficit that stabilizes debt ratio	-14.9	12.2	10.7			23.8	7.9	28.9	16.9	-6.1	-6.3		-5.3	-7.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.1	7.6	8.9	5.0	2.7	7.7	4.0	7.7	20.0	5.4	4.9	8.3	3.0	3.0	3.2
GDP deflator in US dollar terms (change in percent)	-4.5	13.4	17.5	10.0	8.0	13.0	0.7	4.8	14.7	-1.0	-1.0	5.2	1.9	1.9	1.8
Effective interest rate (percent) 5/	4.7	2.8	1.8	3.3	0.8	2.0	1.9	1.9	1.8	1.9	3.7	2.2	6.4	8.9	7.4
Growth of exports of G&S (US dollar terms, in percent)	-22.3	31.1	21.6	14.6	18.1	11.1	3.3	10.0	53.0	3.5	1.0	13.6	2.6	2.6	2.6
Growth of imports of G&S (US dollar terms, in percent)	11.2	48.4	24.4	19.5	15.2	8.8	-7.3	-0.9	-20.4	-1.5	2.6	-3.1	3.0	3.0	3.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	26.9	26.1	24.4	25.6	23.2	23.4	24.9	25.4	25.4	25.2
Government revenues (excluding grants, in percent of GDP)	23.4	26.0	27.3			25.6	24.9	22.6	17.8	17.9	18.3		21.2	22.4	21.2
Aid flows (in Billions of US dollars) 7/	0.3	0.6	0.7			0.4	0.4	0.4	0.4	0.4	0.4		0.4	0.2	
o/w Grants	0.3	0.5	0.3			0.3	0.3	0.3	0.3	0.3	0.3		0.4	0.1	
o/w Concessional loans	0.0	0.1	0.4			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			2.3	2.1	1.9	1.5	1.4	1.4		1.1	0.3	0.8
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			70.0	72.9	72.9	72.2	75.9	76.0		79.6	63.6	74.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	8.1	9.9	12.7			15.4	16.1	18.2	25.0	26.1	27.2		35.1	56.7	
Nominal dollar GDP growth	1.3	22.0	28.0			21.6	4.7	12.9	37.7	4.4	3.9	14.2	4.9	4.9	5.0
PV of PPG external debt (in Billions of US dollars)			1.2			1.3	1.3	1.4	1.5	1.5	1.6		1.8	2.1	
(PVt-PVt-1)/GDPt-1 (in percent)						0.6	0.4	0.4	0.5	0.3	0.3	0.4	0.1	0.0	0.1
Gross workers' remittances (Billions of US dollars)	-0.1	-0.1	-0.2			-0.2	-0.2	-0.2	-0.3	-0.3	-0.3		0.0	-0.2	
PV of PPG external debt (in percent of GDP + remittances)	...	...	8.7			8.5	8.5	8.0	6.1	6.1	6.1		5.3	3.8	
PV of PPG external debt (in percent of exports + remittances)	...	...	14.8			15.9	16.2	15.5	10.7	10.8	11.1		11.0	9.8	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	1.3			1.5	1.4	1.2	0.8	0.7	0.8		0.8	0.8	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes. Large residuals in 2009-14 are due to a difference in timing between LNG-related debt disbursements and the shipping of imported component purchases to PNG. Long-run residuals reflect BPNG reserve accumulation.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table III. 4: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012–2032
(In percent)

	Projections							
	2012	2013	2014	2015	2016	2017	**2022**	2032
PV of debt-to GDP ratio								
Baseline	8	8	8	6	6	6	**5**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	8	-4	-8	0	6	12	**29**	37
A2. New public sector loans on less favorable terms in 2012-2032 2	8	9	8	7	7	7	**7**	6
A3. Strong price decline for Gold, Copper, Oil and Gas from 2013 onwards.	8	12	16	20	29	38	**76**	127
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	8	8	8	6	6	6	**6**	4
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	8	13	22	17	17	17	**14**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	8	9	9	7	7	7	**6**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	8	12	17	13	13	13	**11**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	8	10	20	15	15	15	**13**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	8	12	11	8	8	8	**7**	5
PV of debt-to-exports ratio								
Baseline	16	16	15	11	11	11	**11**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	16	-8	-15	-1	11	21	**60**	93
A2. New public sector loans on less favorable terms in 2012-2032 2	16	16	16	11	12	12	**14**	15
A3. Strong price decline for Gold, Copper, Oil and Gas from 2013 onwards.	15	24	34	43	63	86	**210**	456
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	16	15	15	10	10	11	**11**	9
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	16	27	58	40	40	41	**40**	27
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	16	15	15	10	10	11	**11**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	16	22	33	23	23	24	**23**	16
B5. Combination of B1-B4 using one-half standard deviation shocks	16	20	39	27	27	27	**27**	19
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	16	15	15	10	10	11	**11**	9
PV of debt-to-revenue ratio								
Baseline	33	34	35	34	34	33	**25**	17
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	33	-17	-34	-2	34	65	**137**	165
A2. New public sector loans on less favorable terms in 2012-2032 2	33	34	36	37	37	37	**31**	26
A3. Strong price decline for Gold, Copper, Oil and Gas from 2013 onwards.	32	47	70	114	162	206	**361**	566
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	33	34	37	36	36	35	**27**	18
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	33	52	99	95	94	92	**66**	35
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	33	34	38	37	37	36	**27**	19
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	33	47	77	74	73	72	**52**	29
B5. Combination of B1-B4 using one-half standard deviation shocks	33	42	87	84	83	81	**59**	32
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	33	47	48	47	47	46	**35**	23

Table III. 4: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012–2032 (continued)
(In percent)

Debt service-to-exports ratio								
Baseline	1	1	1	1	1	1	**1**	1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	1	1	1	0	0	1	**1**	4
A2. New public sector loans on less favorable terms in 2012-2032 2	1	1	1	1	1	1	**1**	1
A3. Strong price decline for Gold, Copper, Oil and Gas from 2013 onwards.	1	2	2	1	2	2	**7**	25
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	1	1	1	1	1	1	**1**	1
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	1	2	2	2	2	2	**3**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	1	1	1	1	1	1	**1**	1
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	1	1	1	1	1	1	**2**	1
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	1	1	1	1	**2**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	1	1	1	1	1	1	**1**	1
Debt service-to-revenue ratio								
Baseline	3	3	3	2	2	2	**2**	1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	3	3	1	1	2	2	**3**	8
A2. New public sector loans on less favorable terms in 2012-2032 2	3	3	3	2	2	2	**2**	2
A3. Strong price decline for Gold, Copper, Oil and Gas from 2013 onwards.	3	3	3	3	4	6	**12**	30
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	3	3	3	3	3	2	**2**	1
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	3	3	3	4	4	4	**5**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	3	3	3	3	3	3	**2**	1
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	3	3	3	4	3	3	**4**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	3	3	3	4	4	4	**4**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	3	4	4	3	3	3	**3**	2
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	23	23	23	23	23	23	**23**	23

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.